

their explicit representation, while the preserved sources expose the remaining interpretation questions.

Generalisation requires the same discipline when the next unfamiliar instrument arrives. The program should preserve the new sources before adapting its interpretation, attempt the fixed checks, and report which premises it cannot establish. If a legal change requires repair, the old result remains associated with its original sources and method, and the repaired method is tested on a subsequent untouched case. The defensible promise is conditional correctness with detectable limits and reproducible revision. Universal legal correctness on unknown future texts has not been proved.

2.4 The bank and the transaction also determine eligibility

An ordinary bond can have straightforward payment terms and still be unavailable for a particular transaction. The client may be prohibited from acquiring it; the bank may be prohibited from dealing in it; or the bank may lack information needed to establish which restrictions apply. The CoCo example therefore exposes a general question. After explaining what an instrument promises, the system must establish what the participating institutions and clients may do with it. Changing the instrument from an AT1 bond to a preferred share, a fund or an ordinary equity does not remove that question.

2.4.1 Identify the institution before selecting its rules

Consider a private-banking service provided through a Hong Kong office. JPMorgan's public Asia disclosure identifies JPMorgan Chase Bank, N.A., Hong Kong or Singapore branch as the provider of dealing, advisory, discretionary management, banking and custody services, as notified to the client. It also describes the Hong Kong branch's local regulatory supervision. The same page identifies other legal entities in other jurisdictions. Consequently, the J.P. Morgan brand alone cannot identify the legal person undertaking a trade; the account and service documents must establish that person and its capacity (JPMorgan Chase, 2026, Asia service-provider disclosure).

The US Treasury's Office of Foreign Assets Control (OFAC) includes foreign branches of US entities among US persons. A separately incorporated foreign subsidiary requires a program-specific inquiry: some programs extend obligations to subsidiaries owned or controlled by US persons. The system must preserve that relationship and apply the particular program's definition (Office of Foreign Assets Control, 2024, FAQ 11). For the Chinese securities regime, Executive Order 13959, as amended by Executive Order 14032, includes foreign branches within its US-entity definition, alongside US citizens, lawful permanent residents and persons in the United States. The investor's and the bank's statuses must therefore be recorded separately (President of the United States, 2021, replacement section 3(d)).

The account's booking entity is only the beginning. Advice, execution, custody and

payment processing may involve different entities. Each participant needs an identified role, location and applicable obligations. A decision about an advisory service cannot automatically authorize principal dealing by the bank, and permission for a purchase cannot automatically authorize a subsequent transfer or conversion. These are distinctions the proposed data model must retain even when a particular legal rule ultimately treats two cases alike.

2.4.2 A US intermediary is different from a US investor

The Chinese military-industrial securities restrictions supply a useful counterexample to an overly broad institutional rule. Suppose, conditionally, that the issuer is designated, the security is covered, the restriction has taken effect, and no relevant authorization or divestment exception applies. A purchase by a US person is then prohibited under the amended executive order. Neither Chinese state ownership alone nor an instrument's AT1 label establishes these premises; designation, security coverage and dates require their own evidence ([President of the United States, 2021](#), replacement section 1).

Now change the bank's capacity while preserving those instrument facts. OFAC permits specified investment advisory and management services for a non-US person when the underlying transaction is otherwise permissible. Its guidance explicitly addresses the ultimate benefit of US persons and evasion. This means that a US bank's involvement does not by itself establish that every non-US client transaction is prohibited ([Office of Foreign Assets Control, 2021b](#), FAQ 902). OFAC also states that US financial institutions may continue to intermediate qualifying transactions between non-US persons. Conversely, a prohibited transaction under this particular order must be rejected; the order does not itself require an asset freeze ([Office of Foreign Assets Control, 2022b](#), FAQ 1048). The system therefore needs the actual service and beneficial parties, rather than an undifferentiated "US bank involved" flag.

Time and action introduce another distinction. OFAC permits continued holding after the relevant divestment period, although a purchase or sale after that period requires authorization. An instruction to sell an existing holding can therefore require a different answer from an instruction to retain it ([Office of Foreign Assets Control, 2022a](#), FAQ 1046). The exact prohibition and divestment dates depend on the designation history; an old prospectus cannot supply later list changes ([Office of Foreign Assets Control, 2021a](#), FAQ 899). Automatic conversions, redemptions and other corporate actions require their own analysis. The present conditional implementation supports buying, selling and holding; it leaves an unsupported action unresolved.

Ownership also has program-specific meaning. Under OFAC's blocking ownership rule, suppose blocked person X owns half of company A and A owns half of company B. A becomes blocked through X's ownership, and B becomes blocked through A's ownership. Multiplying the two fractions and treating B as merely one-quarter owned by X misses the applicable propagation. Multiple blocked owners' direct stakes must also be aggregated ([Office of Foreign Assets Control, 2014](#), FAQ 401, examples 1–3). Under the Chinese military-industrial securities regime, however, a subsidiary must itself be

designated: the blocking ownership rule does not apply merely because its parent appears solely on the Chinese military-industrial securities list ([Office of Foreign Assets Control, 2021c, FAQ 857](#)). A reusable ownership graph consequently needs a rule that states which relationships have legal consequences in each program.

2.4.3 Compliance includes institutional and account obligations

Sanctions are one source of additional restrictions. The retained regulation of the Office of the Comptroller of the Currency (OCC) requires national banks to maintain a written Bank Secrecy Act compliance program, with internal controls, independent testing, designated responsibility and personnel training. It also addresses customer identification within that program ([Office of the Comptroller of the Currency, 2025, 12 CFR 21.21\(c\)–\(d\), 2025 edition](#)). These obligations concern the institution's conduct and controls. Reading a security's prospectus cannot establish that the bank has performed them.

Applicability still precedes assessment. In the retained 2025 rules, 31 CFR 1010.620(a) addresses defined private-banking accounts established, maintained, administered or managed in the United States. The account definition adds conditions concerning the required minimum aggregate assets, non-US owners and a liaison with the institution. Thus neither the commercial description "private banking" nor a Hong Kong office address alone decides whether that particular US provision applies. Its substantive requirements include ownership, source-of-funds and account-activity inquiries, with enhanced scrutiny for the specified foreign political figures ([Financial Crimes Enforcement Network, 2025b, 31 CFR 1010.605\(m\), 2025 edition](#)) ([Financial Crimes Enforcement Network, 2025a, 31 CFR 1010.620\(a\)–\(d\), 2025 edition](#)). Those editions illustrate the scope distinction; their preservation does not establish that every provision remains current for a later transaction.

The proposed system separates these institutional and account requirements from product distribution and conduct rules. It also reserves a distinct place for the bank's internal policies and the client's mandate. An internal restricted list or product policy can prevent a service even where the examined public law does not prohibit it. The repository has not received JPMorgan's actual internal policies, so their requirements and permitted exceptions remain missing inputs. They must not be reconstructed from a public marketing disclosure.

Finally, execution and custody require the actual settlement route and relevant reporting duties. A requirement to investigate an account, reject a trade, freeze specified property or make a report has a different legal effect. The decision must preserve those distinctions. A generic negative eligibility flag cannot safely stand for all four actions, and satisfying one duty does not establish that all the others have been performed.

2.4.4 Compose the obligations without erasing qualifications

The common front end now needs to supply two kinds of information together: the instrument's sourced terms and the transaction's parties, institutions, services and dates. Rule packs selected for that context supply the applicable requirements. In the Python implementation these are organized into seven assessment categories: scope and coverage; sanctions; bank regulation and licensing; client and account controls; product distribution and conduct; bank policy and mandate; and settlement, custody and reporting. The categories are a software design choice. They do not assert that every relevant rule has already been discovered or implemented.

Each required assessment carries a result, a reason, source references and a recorded validity interval. A finding that a rule does not apply needs evidence just as a finding that its requirements are satisfied does. Unknown, conflicting, expired and missing findings prevent an affirmative result. The binding also identifies the instrument, client, booking entity, establishment, service, action, transaction time, knowledge time, and content hashes for the facts, settlement route and policy snapshot. Changing any of them invalidates the earlier bound finding. Checking a source's hash establishes unchanged bytes; the scope assessment must separately address legal currentness and meaning.

The conjunction has a simple conditional justification. Let R be the finite set of declared requirements. Let S contain requirements supported as satisfied or inapplicable by valid, consistent findings for this context. Let U contain requirements with unresolved findings, and let C mean that every required assessment category is represented. The implementation's permission within that declared scope is

$$P = C \wedge (R \subseteq S) \wedge (U = \emptyset).$$

A valid prohibition excludes its requirement from S , and a conflicting favorable finding also places that requirement in U . Either prevents P . An absent bank-policy assessment has the same practical consequence for permission, but its explanation is missing evidence rather than a proved legal prohibition. A favorable product assessment cannot alter either result.

This composition is independent of the final rule language. The intended RuleIR and Catala translations must consume the same facts, applicability premises and rule versions. At present the added component is Python: its selected sanctions calculations and seven-category composition have not yet been lowered into both native backends. The earlier backend agreement on product rules consequently supplies no new evidence for these sanctions rules.

The checks enumerate every combination of satisfied, prohibited and unknown across the seven categories, giving $3^7 = 2,187$ cases. Further cases challenge conflicts, non-applicability, missing policy, source changes and changed entity, action or time. The ownership calculation has an independent finite check: for each of 512 three-entity graph and initial-blocking combinations, the test intersects every ownership-closed superset of the initial blocked entities and compares that set with the implementation's result. The algorithm is monotone and terminates on a finite graph; induction shows

that every closed superset contains each entity it adds. Its terminal set is therefore the least closed superset under the supplied ownership premises.

No human quality labels are used in these checks. They establish properties of the declared formal rules and the handling of missing evidence. Correctness of a new legal interpretation, completeness of an ownership register and coverage of future amendments remain separate obligations. The system can generalize by applying checked semantics to new supported facts and qualifying unsupported situations; it cannot turn today's finite legal collection into a proof about every law that might govern a future transaction.

2.5 Marketing rules require exceptions and judgment

The October 2023 distributor circular examines additional returns and incentives, redemption restrictions and marketing of fund-related services. Its treatment of guaranteed-return arrangements depends on the arrangement and the relevant legal classification. A potential structured-product classification should produce a supported assessment task; a language model should not resolve it from a slogan alone (SFC, 2023b, paras. 5–11).

The circular's discussion of gifts expressly distinguishes discounts of fees or charges. Losing that exception makes an apparently conservative implementation incorrectly broad. Its redemption discussion distinguishes administrative procedures or cut-off times from reducing a fund's dealing frequency. A daily dealing fund offered with weekly redemption presents a different issue from a different daily processing cut-off. These are excellent paired test cases because they differ in the fact that determines the treatment (SFC, 2023b, paras. 9–14).

The advertising discussion addresses misleading impressions and comparisons with deposits. Keyword detection can nominate material for review, but it cannot establish the whole classification: meaning depends on the presentation and the service described. The system should retain the advertisement, the relevant source passages and the reviewer's conclusion. That is a useful result even when there is no defensible fully automated predicate (SFC, 2023b, paras. 15–17).

2.6 The source packet and the question it supports

The running example begins with an identifiable public document: SFC circular 23EC46, issued on 24 October 2023, concerning additional returns and other services or arrangements offered when marketing SFC-authorized funds (SFC, 2023b). The retained experiment selected its paragraph-10 gift restriction. The circular as a whole discusses more than that restriction. A reader needs to know both what was selected and what remains outside the control before a result from the control can be interpreted.